

STA258H5

Statistics with Applied Probability

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SAMPLING DISTRIBUTIONS RELATED TO A NORMAL POPULATION

Theorem 7.1

$$\text{var}(cx) = c^2 \text{var}(x) \quad \text{var}\left(\frac{\bar{X}}{\sigma/\sqrt{n}}\right) = \frac{1}{\sigma^2/n} \cdot \text{var}(\bar{X}) = \frac{1}{\sigma^2/n} \cdot \frac{\sigma^2}{n}$$

Let $Y_1, Y_2, \dots, Y_n$ be a random sample of size n from a Normal distribution with mean μ and variance σ^2 . Then

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

is Normally distributed with mean $\mu_{\bar{Y}} = \mu$ and variance $\sigma_{\bar{Y}}^2 = \frac{\sigma^2}{n}$.

$$\text{(or } Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)) \quad \bar{X} \sim N(\mu_{\bar{X}} = \mu, \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n})$$

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(\mu_{\bar{X}} = 0, \sigma_{\bar{X}}^2 = 1)$$

Example

$$\mu = 75 \quad \sigma = 15 \quad n = 30$$
$$\bar{x} \sim N\left(\mu_{\bar{x}} = 75, \sigma_{\bar{x}}^2 = \frac{15^2}{30}\right)$$
$$\sigma_{\bar{x}} = \sqrt{\sigma^2} = \frac{15}{\sqrt{30}} = 2.738$$

Marks on a standardized test are Normally distributed with mean 75, standard deviation 15. What is the probability the class average, for a class of 30, is greater than 76?

$\bar{x}$

$$P(\bar{x} > 76)$$

$$= P\left(z > \frac{76 - 75}{2.738}\right)$$

$$= P(z > 0.37) = 0.3557$$

It can be shown that χ_1^2 distribution is the Gamma($\alpha = 1/2, \beta = 2$) distribution.

$$f(y) = \begin{cases} \frac{1}{\Gamma(1/2)2^{1/2}} y^{1/2-1} e^{-y/2} & y > 0 \\ 0 & y \leq 0 \end{cases}$$

$$E(Y) = 1$$

$$Var(Y) = 2.$$

Example

Let Z be a Normally distributed random variable with mean 0 and variance 1. Use the method of moment-generating functions to find the probability distribution of Z^2 .

Solution

$$\begin{aligned} M_{Z^2}(t) &= E(e^{tZ^2}) = \int_{-\infty}^{\infty} e^{tz^2} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2(\frac{1-2t}{2})} dz \end{aligned}$$

This integral can be evaluated using an “old trick” (we note that it looks like a Normally distributed random variable).

Solution

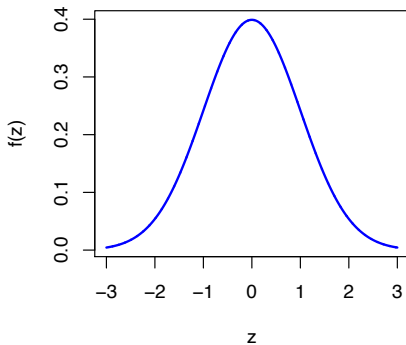
We realize that $e^{-z^2(\frac{1-2t}{2})}$ is proportional to a Normal with $\mu = 0$ and $\sigma^2 = 1/(1-2t)$, then

$$M_{Z^2}(t) = \frac{\sqrt{2\pi}\sqrt{1/(1-2t)}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sqrt{1/(1-2t)}} e^{-z^2(\frac{1-2t}{2})} dz$$

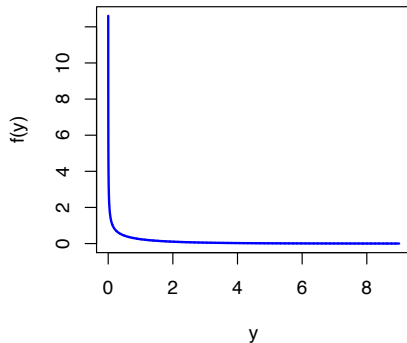
$$M_{Z^2}(t) = \sqrt{\frac{1}{1-2t}} = (1-2t)^{-1/2} \text{ (Note. This is valid provided that } t < 1/2\text{).}$$

$(1-2t)^{-1/2}$ is the moment-generating function for a gamma-distributed random variable with $\alpha = 1/2$ and $\beta = 2$. Hence, Z^2 has a χ^2 distribution with $\nu = 1$ degree of freedom.

N(0,1) distribution



Chi-square dist, df=1



Example

Suppose that Y_1 and Y_2 are independent, standard Normal random variables. Find the probability distribution of $U = Y_1^2 + Y_2^2$.

$$\begin{aligned}M_U(t) &= E[e^{Ut}] = E[e^{(Y_1^2 + Y_2^2)t}] \\&= E[e^{Y_1^2 t}]E[e^{Y_2^2 t}] \quad (Y_1 \text{ and } Y_2 \text{ are independent}) \\&= M_{Y_1^2}(t)M_{Y_2^2}(t) \\&= [(1 - 2t)^{-1/2}][(1 - 2t)^{-1/2}] = (1 - 2t)^{-2/2}.\end{aligned}$$

Because moment-generating functions are unique, U has a χ^2 distribution with 2 degrees of freedom.

χ^2 Distribution

χ Chi (pronounced Kai)

cts distrib. Density fn is messy.

Shape changes with n (degrees of freedom)

Let $z_1, z_2, \dots, z_n \stackrel{\text{iid}}{\sim} N(0, 1)$

The distribution of

$$z_1^2 + z_2^2 + \dots + z_n^2 \sim \chi_n^2$$



proof $\rightarrow$ STA 260
(uses MGF)

A chi-sqr distrib
with n df is the
distrib of the
sum of n
standard normal
variables squared

$$M_{z_1^2 + \dots + z_n^2}(t) = M_{z_1^2}(t) \cdot \dots \cdot M_{z_n^2}(t)$$

A result of this (w/o proof) :

$$\text{Let } s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2 \quad (\text{Sample variance})$$

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

$$\downarrow$$
$$s^2 \sim \frac{\sigma^2}{(n-1)} \chi^2_{(n-1)} \quad \left. \vphantom{\frac{\sigma^2}{(n-1)} \chi^2_{(n-1)}} \right\} \text{ Scaled } \chi^2$$

Properties of χ_n^2

$$\text{Let } X \sim \chi_n^2$$

$$\text{mean} : E(X) = n$$

$$\text{var} : \text{var}(X) = 2n$$

$$\text{Let } V = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

$$E(V) = n-1, \quad \text{Var}(V) = 2(n-1)$$

$$\text{Let } S^2 \sim \frac{\sigma^2}{(n-1)} \chi^2_{(n-1)}$$

$$E(gx) = g E(x)$$

$$\begin{aligned} E(S^2) &= E\left[\frac{\sigma^2}{n-1} \chi^2_{(n-1)}\right] = \frac{\sigma^2}{n-1} \cdot E(\chi^2_{n-1}) \\ &= \frac{\sigma^2}{n-1} \cdot n-1 \quad \checkmark \\ &= \sigma^2 \end{aligned}$$

$$\text{var}(S^2) = \text{var}\left[\frac{\sigma^2}{n-1} \chi^2_{(n-1)}\right]$$

$$\text{var}(aX) = a^2 \text{var}(X)$$

$$= \left(\frac{\sigma^2}{n-1}\right)^2 \cdot \text{var}(\chi^2_{n-1})$$

$$= \frac{\sigma^4}{(n-1)^2} \cdot 2(n-1)$$

$$= \frac{2\sigma^4}{n-1}$$

Definition. Let $Z_1, Z_2, \dots, Z_n$ be iid standard Normal random variables. Define $V = \sum_{i=1}^n Z_i^2$. V has a χ_n^2 distribution. The χ_n^2 is the Gamma($\alpha = n/2, \beta = 2$) distribution.

Proof.

By uniqueness of moment generating functions.

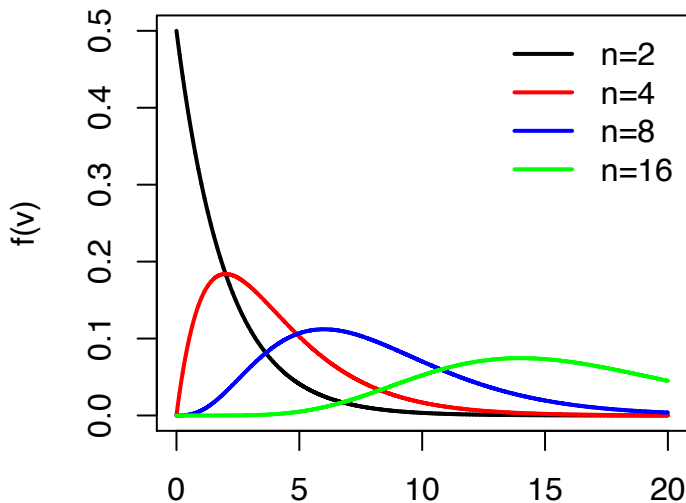
χ_n^2 distribution

$$f(v) = \begin{cases} \frac{1}{\Gamma(n/2)2^{n/2}} v^{n/2-1} e^{-v/2} & v > 0 \\ 0 & v \leq 0 \end{cases}$$

$$E(V) = n \qquad \text{Var}(V) = 2n.$$

The subscript n is called the degrees of freedom; it is the number of “free” variables in the sum.

Chi-square distributions



Theorem 7.2

Let $Y_1, Y_2, \dots, Y_n$ be defined as in Theorem 7.1. Then $Z_i = \frac{Y_i - \mu}{\sigma}$ are independent, standard Normal random variables, $i = 1, 2, \dots, n$, and

$$\sum_{i=1}^n Z_i^2 = \sum_{i=1}^n \left(\frac{Y_i - \mu}{\sigma} \right)^2$$

has a χ^2 distribution with n degrees of freedom (df).

It can be shown (see STA260) that

If $X_1, X_2, \dots, X_n \sim \text{iid } N(\mu, \sigma^2)$, then $V = \sum_{j=1}^n \left(\frac{X_j - \bar{X}}{\sigma} \right)^2 \sim \chi_{n-1}^2$

Theorem 7.3

Let $Y_1, Y_2, \dots, Y_n$ be a random sample from a Normal distribution with mean μ and variance σ^2 . Then

$$\frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

has a χ^2 distribution with $(n-1)$ df.

Also, $\bar{Y}$ and S^2 are independent random variables.

Since, if $Y_1, Y_2, \dots, Y_n \sim \text{iid } N(\mu, \sigma)$, then $V = \sum_{i=1}^n \left(\frac{Y_i - \bar{Y}}{\sigma} \right)^2 \sim \chi_{n-1}^2$

It follows that, if $Y_1, Y_2, \dots, Y_n \sim \text{iid } N(\mu, \sigma)$ and

$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$, then $V = \sum_{i=1}^n \left(\frac{Y_i - \bar{Y}}{\sigma} \right)^2 = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$.

$E(S^2) = \sigma^2$ and $Var(S^2) = \frac{2\sigma^4}{n-1}$.

Example



The Environmental Protection Agency is concerned with the problem of setting criteria for the amounts of certain toxic chemicals to be allowed in freshwater lakes and rivers. A common measure of toxicity for any pollutant is the concentration of the pollutant that will kill half of the test species in a given amount of time (usually 96 hours for fish species). This measure is called LC50 (lethal concentration killing 50% of test species). In many studies, the values contained in the natural logarithm of LC50 measurements are Normally distributed, and, hence, the analysis is based on $\ln(\text{LC50})$ data.

Example

Suppose that $n = 20$ observations are to be taken on $\ln(\text{LC50})$ measurements and that $\sigma^2 = 1.4$. Let S^2 denote the sample variance of the 20 measurements.

- Find a number b such that $P(S^2 \leq b) = 0.975$.
- Find a number a such that $P(a \leq S^2) = 0.975$.
- If a and b are as in parts a) and b), what is $P(a \leq S^2 \leq b)$?

Slide 20

(Thm 7.3)

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

Find b s.t.

$$P(s^2 \leq b) = 0.975$$

$$P\left(\underbrace{\frac{(n-1)s^2}{\sigma^2}}_{\chi^2_{n-1}} \leq \frac{(n-1)b}{\sigma^2}\right) = 0.975$$

$$\begin{aligned} n &= 20 \\ \sigma^2 &= 1.4 \end{aligned}$$

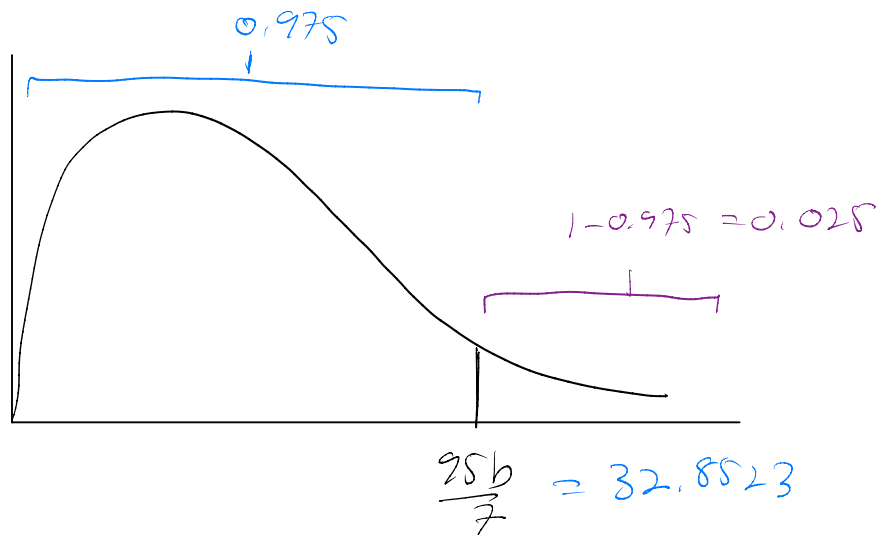
$$P\left(\chi^2_{n-1} \leq \frac{(20-1) \cdot b}{1.4}\right) = 0.975$$

$$b = ?$$

$$P\left(\chi^2_{n-1} \leq \frac{95b}{7}\right) = 0.975$$

$$n-1 = 20-1 = 19 \text{ df}$$

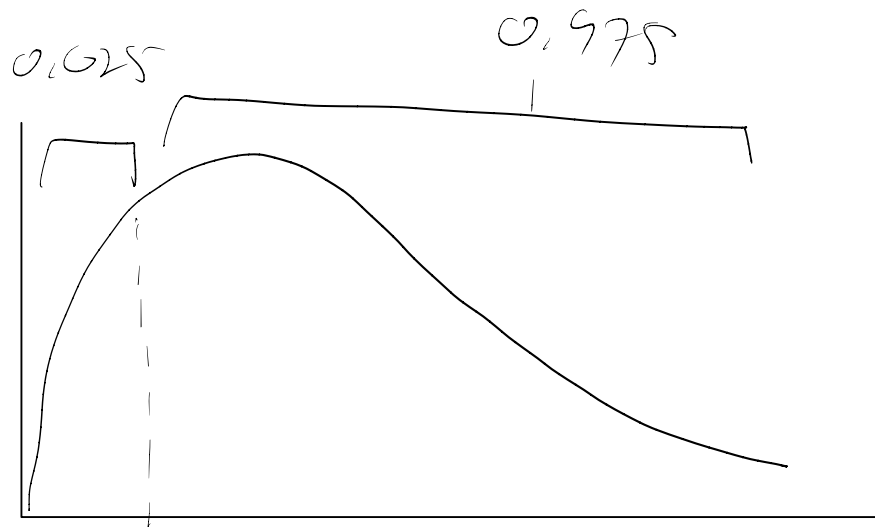
using $\chi^2_{19} \leftarrow \text{df}$



$$\downarrow$$
$$\boxed{b = 2.42}$$

b) Find a such that $P(a \leq S^2) = 0.975$

4
5
7
!



$$\frac{95a}{7} = 8.9065$$

$$a = 0.656$$

exercise

$$c) P(a \leq S^2 \leq b) = ?$$

$$P(8.9065 \leq \underbrace{\frac{(n-1)S^2}{\sigma^2}}_{\sim \chi^2_{19}} \leq 32.852)$$

$$= 0.975 - 0.025 = 0.95$$

These values can be found by using percentiles from the chi-square distribution.

With $\sigma^2 = 1.4$ and $n = 20$,

$\frac{n-1}{\sigma^2} S^2 = \frac{19}{1.4} S^2$ has a chi-square distribution with 19 degrees of freedom.

$$\text{a. } P(S^2 \leq b) = P\left(\frac{n-1}{\sigma^2} S^2 \leq \frac{(n-1)b}{\sigma^2}\right) = P\left(\frac{19}{1.4} S^2 \leq \frac{19b}{1.4}\right) = 0.975$$

$\frac{19b}{1.4}$ must be equal to the 97.5%-tile of a chi-square with 19 df, thus $\frac{19b}{1.4} = 32.8523$ (using Table 6). And so, $b = 2.42$

b. Similarly, $P(S^2 \geq a) = P\left(\frac{n-1}{\sigma^2} S^2 \geq \frac{(n-1)a}{\sigma^2}\right) = 0.975$. Thus, $\frac{19a}{1.4} = 8.90655$, the 2.5%-tile of this chi-square distribution, and so $a = 0.656$.

c. $P(a \leq S^2 \leq b) = P(0.656 \leq S^2 \leq 2.42) = 0.95$.

Example

Let X equal the weight (in grams) of a nail of the type that is used for making decks. Assume that the distribution of X is $N(\mu = 8.78, \sigma^2 = 0.16)$. Let S^2 be the sample variance of the nine weights. Find constants a and b so that $P[a \leq S^2 \leq b] = 0.90$.

Solution (using R)

```
b=qchisq(0.95,df=8)*0.16/8;
```

```
b
```

```
## [1] 0.3101463
```

```
a=qchisq(0.05,df=8)*0.16/8;
```

```
a
```

```
## [1] 0.05465274
```

pchisq()

Marks on a standardized test are Normally distributed with mean 75, standard deviation 15. What is the probability the class sample standard deviation, for a class of 31, is greater than 16?

Definition 7.2: Student's t distribution

Let Guinness

$$T = \frac{Z}{\sqrt{W/r}},$$

distrib for means
when pop variance
is not known

where Z is a random variable that is $N(0, 1)$, W is a random variable that is $\chi^2(r)$, and Z and W are independent. Then T has a t distribution with r degrees of freedom, for which the pdf is

$$f(t) = \frac{\Gamma((r+1)/2)}{\sqrt{\pi r} \Gamma(r/2)} \frac{1}{(1 + t^2/r)^{(r+1)/2}}, -\infty < t < \infty.$$

Gosset



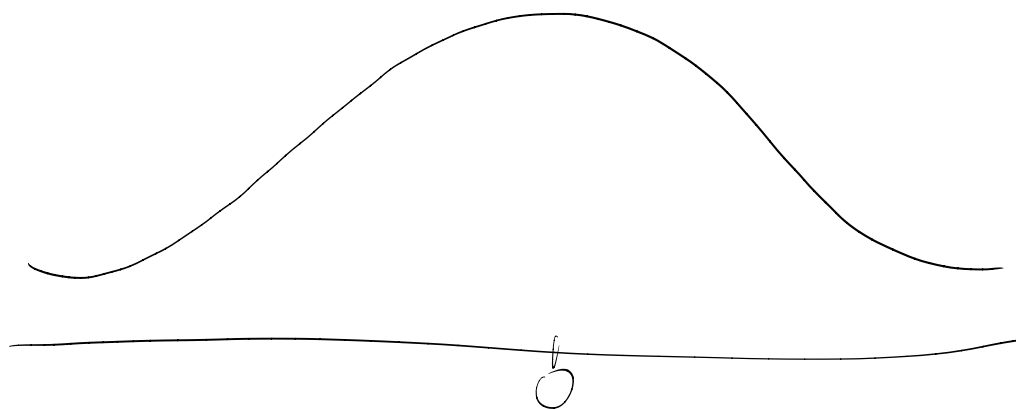
Secretly published
as 'Student'

t-distrib

A t random variable is obtained from the ratio of a normal RV and the square root of a χ^2_r divided by r.

$$t = \frac{Z \sim N(0,1)}{\sqrt{\frac{W}{r}} \sim \chi^2_r} \sim t_r$$

Similar in shape to the standard normal distribution but with thicker tails.



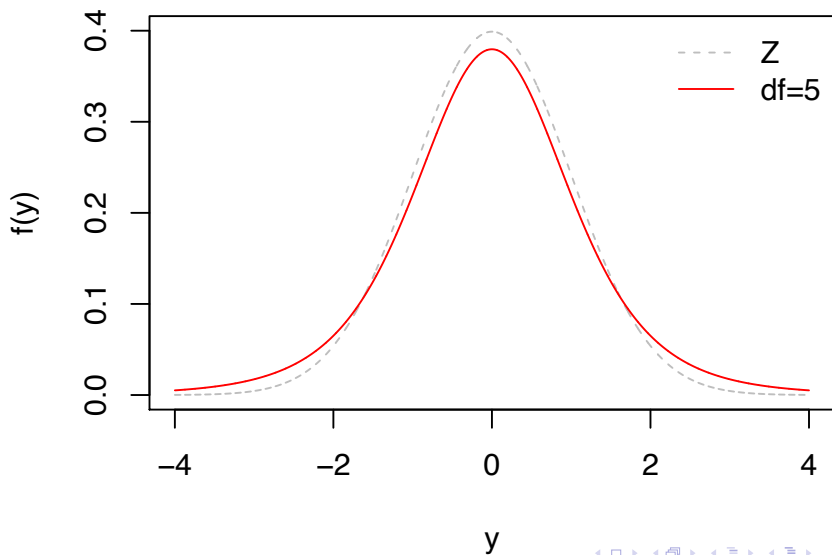
as $df \rightarrow \infty$, $t \rightarrow Z$ mean = 0

$\text{var}(t)$: $df > 2$ $\text{var} = \frac{df}{df-2}$

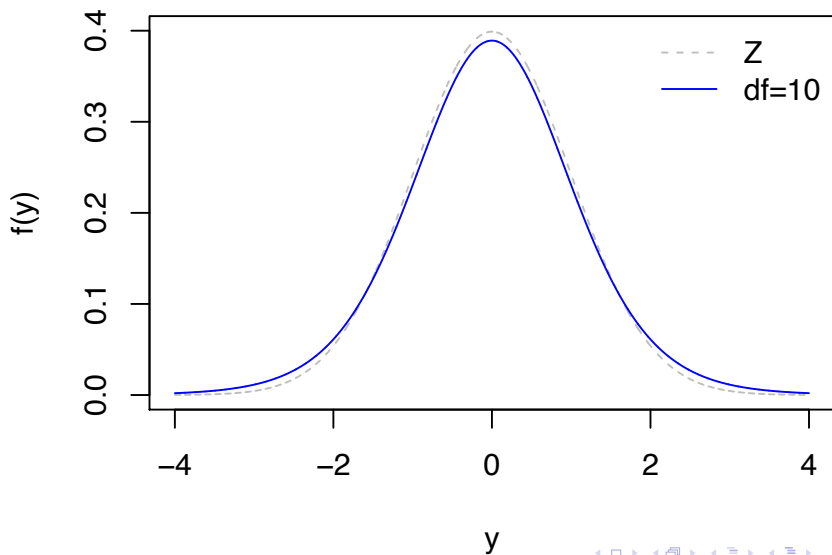
$1 \leq df \leq 2$ $\text{var} \rightarrow \infty$

$df < 1$ var is undef

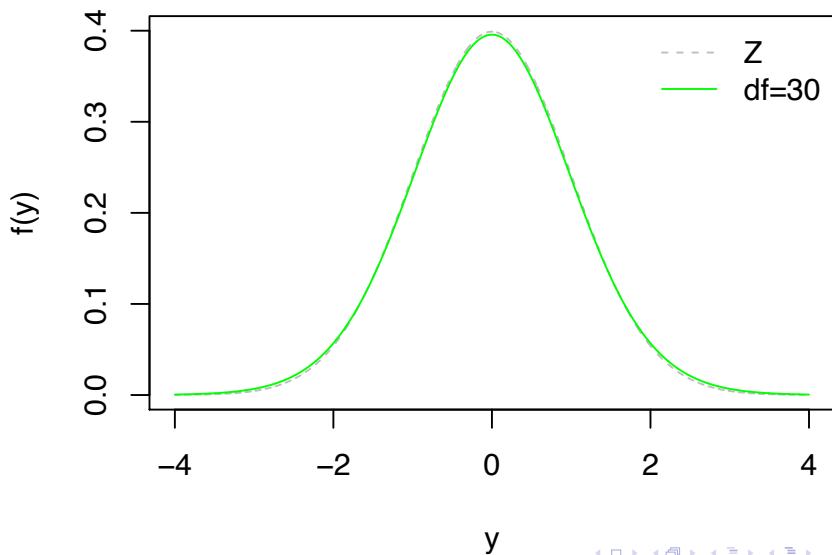
t distributions



t distributions



t distributions



Properties of the t distribution, $T \sim t_\nu$

- Support - the real numbers
- $E(T) = 0$
- $Var(T) = \frac{\nu}{\nu-2} > 1$ $Var(T) \rightarrow 1$ as $\nu \rightarrow \infty$
- The t_ν is shorter and fatter than the $N(0, 1)$ distribution. The tails of the t_ν decrease to zero slower than those for the $N(0, 1)$.
- $t_\nu \rightarrow Z$ (in distribution) as $\nu \rightarrow \infty$

Question: Why is the t_ν wider and more tail heavy than the $N(0, 1)$.

Answer:

$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$ contains variability from $\bar{X}$.

$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$ contains variability from $\bar{X}$ and from S .

S is sample st. dev

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

(see again for inference)

use when pop var σ is not known

Very important use of the t distribution

If $Y_1, Y_2, \dots, Y_n \sim \text{iid } N(\mu, \sigma^2)$, then

① $\bar{Y} \sim N(\mu, \sigma^2/n)$ or $Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$

② $\frac{n-1}{\sigma^2} S^2 \sim \chi_{n-1}^2$

③ $\bar{X}$ and S^2 are independent (see STA260)

Thus $\frac{\frac{\bar{Y} - \mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{n-1}{\sigma^2} S^2 / (n-1)}} = \frac{\bar{Y} - \mu}{S/\sqrt{n}} \sim t_{n-1}$

!

Example

$$n=15$$

Modify

$$\bar{X}=290$$

$$S=20 \text{ (Not } \sigma)$$

std dev of sample
of 15

The Acme Corporation manufactures light bulbs. The CEO claims that an average Acme light bulb lasts 300 days. A researcher randomly selects 15 bulbs for testing. (The sampled bulbs last an average of 290 days, with a standard deviation of ~~50~~²⁰ days.) If the CEO's claim were true, what is the probability that 15 randomly selected bulbs would have an average life of no more than 290 days?

Note. Please, assume Normality.

Careful with wording

$S=20$
is calculated
from sample

Remember

pop. st. dev σ known

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

pop. st. dev σ NOT known

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

Example (slide 33)

$$n=15, \bar{x}=290,$$

sample st. dev
 $S=20$

Find prob $\bar{x}$ is ≤ 290

$$P(\bar{x} \leq 290)$$

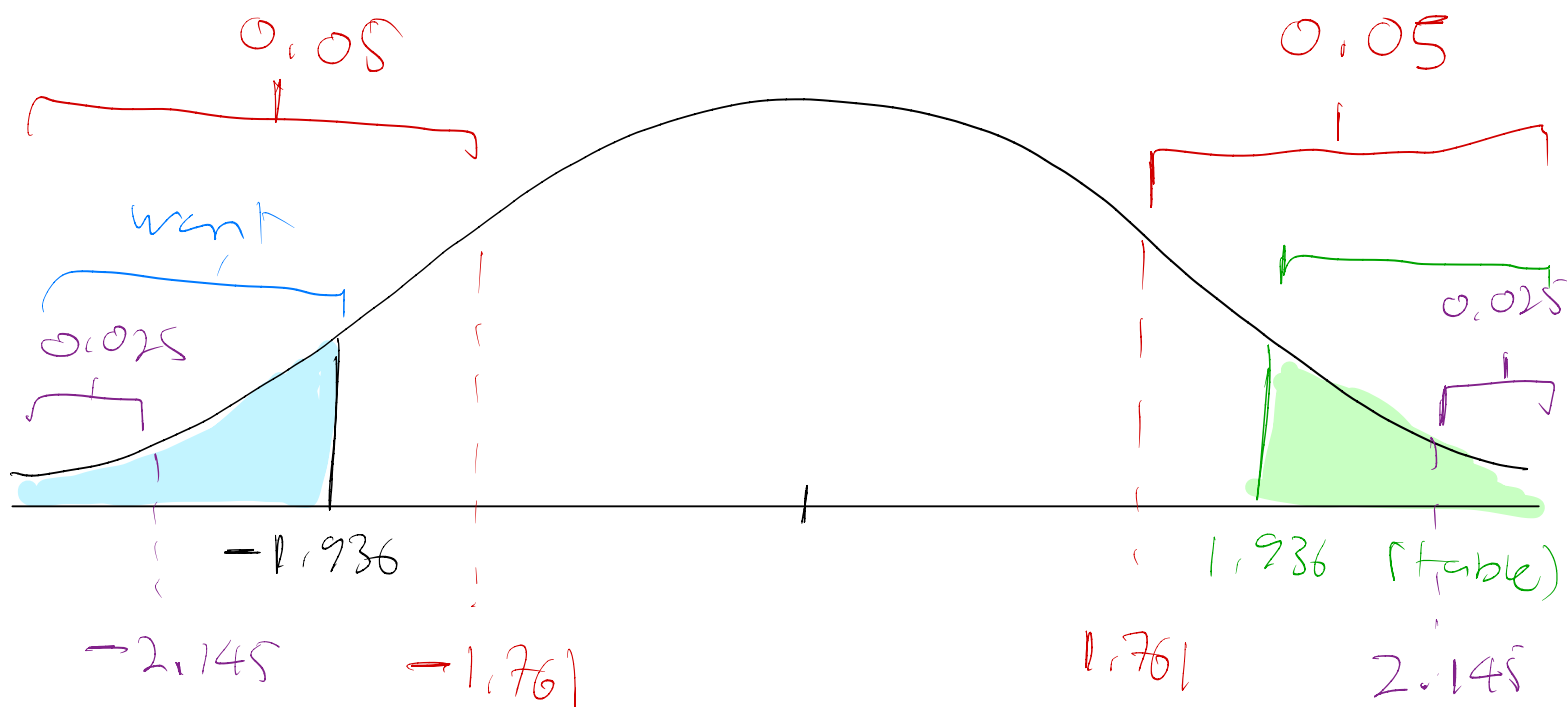
$$= P\left(\underbrace{\frac{\bar{x} - \mu}{S/\sqrt{n}}}_{t_{n-1}} \leq \frac{290 - \mu}{S/\sqrt{n}} \right)$$

$$= P\left(t_{14} \leq \frac{290 - \overset{\text{claimed } \mu}{300}}{20/\sqrt{15}} \right)$$

$$\begin{aligned} df &= n-1 \\ &= 15-1 \\ &= 14 \end{aligned}$$

$$= P(t_{14} \leq -1.936)$$

t at 14 df



want

$$0.025 < p(t_{14} \leq -1.936) < 0.05$$

best we can do with table

$X_1, X_2, \dots, X_{15}$ iid random variables from a $N(\mu = 300, \sigma)$.

$$\begin{aligned} P(\bar{X} \leq 290) &= P\left(\frac{\bar{X} - \mu}{s/\sqrt{n}} \leq \frac{290 - 300}{50/\sqrt{15}}\right) \\ &= P(t_{14} \leq -0.7745) \text{ (since the } t \text{ distribution is symmetric)} \\ &= P(t_{14} \geq 0.7745) \text{ (using our } t \text{ distribution table)} \end{aligned}$$

The best that we can do using table is:

$$0.20 < P(\bar{X} \leq 290) < 0.25$$


```
# pt = CDF of a ttf distribution;
```

```
pt(-0.7745, 14);
```

```
## [1] 0.2257589
```

Example

The Edison Electric Institute (EEI) has published figures on the annual number of kilowatt-hours expended by various home appliances. It is claimed that a vacuum cleaner expends an average of 46 kilowatt-hours per year. A random sample of 12 homes included in a planned study indicates that vacuum cleaners expend an average of 42 kilowatt-hours per year with a standard deviation of 11.9 kilowatt-hours.

If the EEI's claim were true, what is the probability that 12 randomly selected vacuum cleaners expend, on the average, less than 42 kilowatt-hours annually? Assume the population of kilowatt-hours to be Normal.

Solution

$X_1, X_2, \dots, X_{12}$ iid random variables from a $N(\mu = 46, \sigma)$.

$$\begin{aligned} P(\bar{X} \leq 42) &= P\left(\frac{\bar{X} - \mu}{s/\sqrt{n}} \leq \frac{42 - 46}{11.9/\sqrt{12}}\right) \\ &= P(t_{11} \leq -1.1644) \text{ (since the } t \text{ distribution is symmetric)} \\ &= P(t_{11} \geq 1.1644) \text{ (using our } t \text{ distribution table)} \end{aligned}$$

The best that we can do using table from textbook is:

$$P(\bar{X} \leq 42) > 0.10$$


```
# pt = CDF of a ttf distribution;
```

```
pt(-1.1644 , 11);
```

```
## [1] 0.1344472
```

Definition 7.3

Let W_1 and W_2 be independent χ^2 -distributed random variables with ν_1 and ν_2 df, respectively. Then

$$F = \frac{W_1/\nu_1}{W_2/\nu_2} \sim F_{\nu_1, \nu_2}$$

is said to have an F distribution with ν_1 numerator degrees of freedom and ν_2 denominator degrees of freedom.



Ronald Fisher

F-Distribution (Snedecore and Fisher)

Ratio between 2 χ^2 Rv's and their df's

An F distrib has 2 df's (numerator df, denominator df)

$$F = \frac{\chi_m^2 / m}{\chi_n^2 / n}$$

numerator
denominator
down

$$F \sim F(m, n)$$

num df denum df

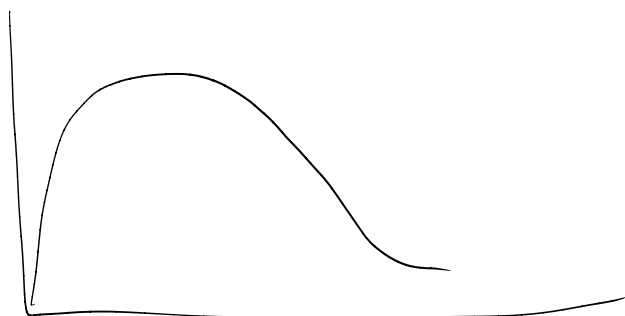
Important result (follows from χ^2 result)

$$F = \frac{\overbrace{\left(\frac{n-1}{\sigma^2} \right) s_1^2}^{\chi_{n-1}^2} / (n-1)}{\underbrace{\left(\frac{m-1}{\sigma^2} \right) s_2^2}_{\chi_{m-1}^2} / (m-1)}$$

Sample from pop 1
has var σ

$$\sim F_{(n-1), (m-1)}$$

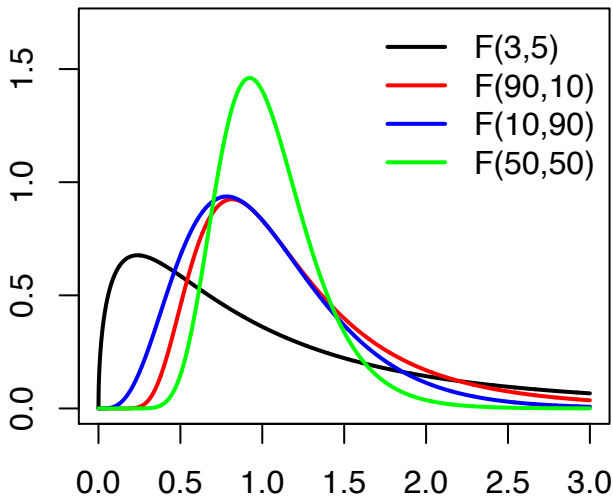
Sample from pop 2
has var σ



$$P_{\text{f}}(\text{---}, \text{num df}, \text{denom df})$$

Usage of F! (For now) compare ratio of Sample variances

F distributions



Properties of F distribution, F_{ν_1, ν_2}

- Support - non-negative real numbers
- Relationship with its reciprocal: If $F \sim F_{\nu_1, \nu_2}$ then $W_1 = 1/F \sim F_{\nu_2, \nu_1}$
- Relationship with the t distribution: If $T \sim t_\nu$ then $W_2 = T^2 \sim F_{1, \nu}$
- Relationship with the exponential distribution: If U_1 and U_2 are independent exponential RVs with the same parameter, then $W_3 = U_1/U_2 \sim F_{2, 2}$

Very important use of the F distribution

If

- ① $X_1, X_2, \dots, X_n \sim \text{iid } N(\mu_X, \sigma^2)$, then $\frac{n-1}{\sigma^2} S_X^2 \sim \chi_{n-1}^2$
- ② $Y_1, Y_2, \dots, Y_m \sim \text{iid } N(\mu_Y, \sigma^2)$, then $\frac{m-1}{\sigma^2} S_Y^2 \sim \chi_{m-1}^2$
- ③ X s and Y s all independent

then

$$F = \frac{\left(\frac{n-1}{\sigma^2} S_X^2\right) / (n-1)}{\left(\frac{m-1}{\sigma^2} S_Y^2\right) / (m-1)} = \frac{S_X^2}{S_Y^2} \sim F_{n-1, m-1}$$

Example

Marks on a standardized test are normally distributed with mean 75, standard deviation 15. The test is given to two classes, one class (A) has 31 students and the other (B) has 16 students.

- What is the probability the ratio of the two sample variances (A/B) is greater 2?
- What is the probability the ratio of the two sample variances (B/A) is less than 2?

class A
 $n_A = 31$ $\sigma^2 = 15^2$

class B
 $n_B = 16$ $\sigma^2 = 15^2$

common variance

ratio of $(S_A^2 / S_B^2) > 2$

$$P\left(\frac{S_A^2}{S_B^2} > 2\right)$$

$$P\left(\frac{\frac{(n_A-1) \cdot S_A^2}{(n_A-1) \sigma^2}}{\frac{(n_B-1) \cdot S_B^2}{(n_B-1) \sigma^2}} > 2\right)$$

$$P\left(\frac{\frac{\chi^2_{n_A-1} \cdot S_A^2}{\sigma^2}}{\frac{\chi^2_{n_B-1} \cdot S_B^2}{\sigma^2}} > 2\right)$$

$$P\left(\frac{\chi^2_{n_A-1} / n_A - 1}{\chi^2_{n_B-1} / n_B - 1} > 2\right)$$

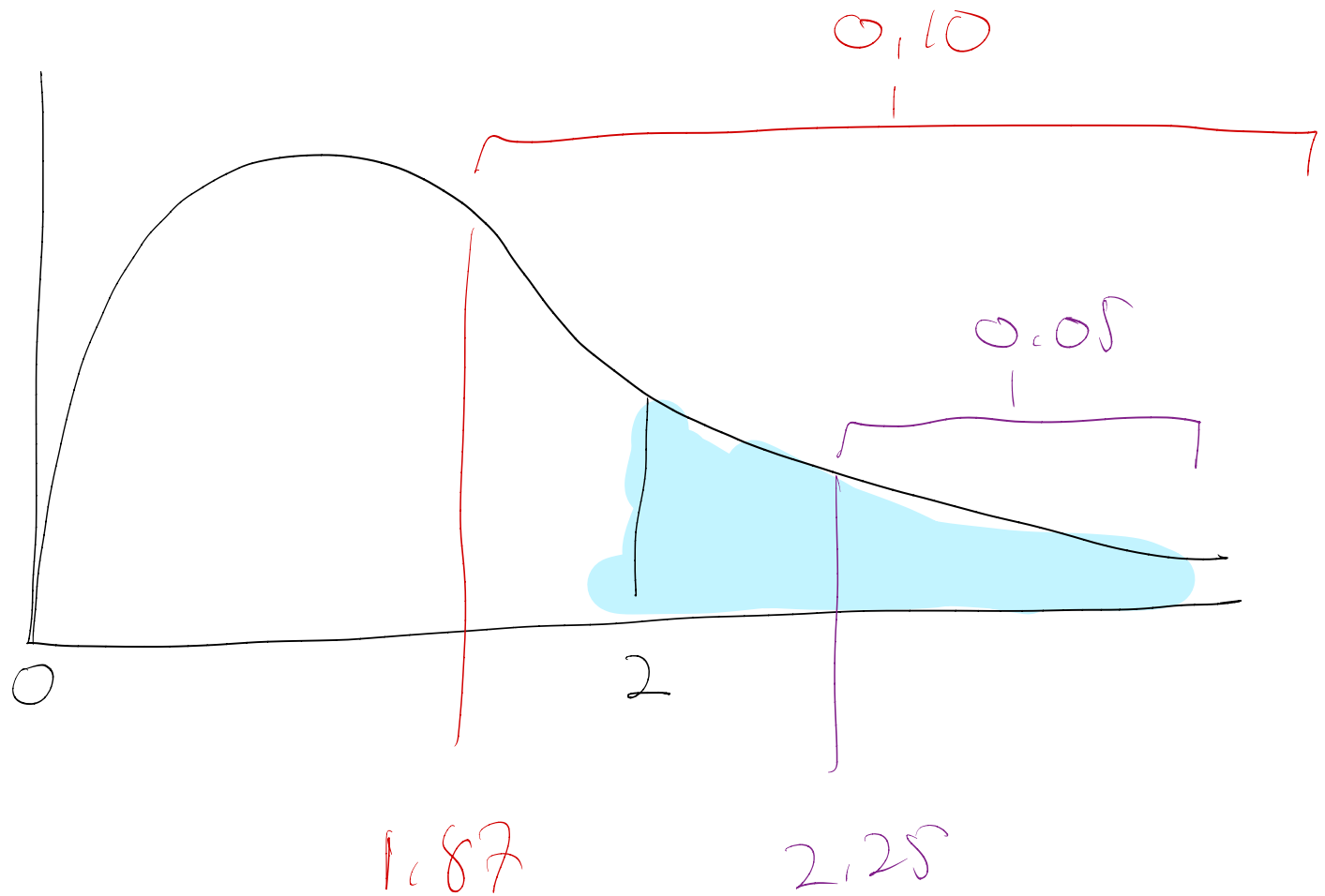
$F_{(n_A-1, n_B-1)}$

$$n_A = 31$$

$$n_B = 15$$

$$P(F_{(30,14)} > 2)$$

using an F at num = 30, denom = 14 df



Best is

$$0.05 < P(F_{(30,14)} > 2) < 0.10$$

Solution (A/B)

$$\mu = 75, \sigma = 15, n = 31, m = 16.$$

$$\begin{aligned} P\left(\frac{S_X^2}{S_Y^2} > 2\right) &= P\left(\frac{\sigma^2}{\sigma^2} \frac{S_X^2}{S_Y^2} > 2\right) \\ &= P\left(\frac{S_X^2/\sigma^2}{S_Y^2/\sigma^2} > 2\right) \\ &= P\left(\frac{\frac{n-1}{\sigma^2} \frac{S_X^2}{n-1}}{\frac{m-1}{\sigma^2} \frac{S_Y^2}{m-1}} > 2\right) \\ &= P\left(\frac{\chi_{n-1}^2/(n-1)}{\chi_{m-1}^2/(m-1)} > 2\right) \\ &= P(F_{n-1, m-1} > 2) = P(F_{30,15} > 2) \end{aligned}$$

Using our Table, the best that we can do is:

$$0.05 < P\left(\frac{S_X^2}{S_Y^2} > 2\right) < 0.10.$$

```
# pf = CDF of F ;  
  
# area to the left of 2;  
  
pf(2,30,15);  
  
## [1] 0.9213152  
  
# area to the right of 2;  
  
1-pf(2,30,15);  
  
## [1] 0.07868481
```

Solution (B/A)

$\mu = 75, \sigma = 15, n = 31, m = 16.$

$$\begin{aligned} P\left(\frac{S_Y^2}{S_X^2} > 2\right) &= P\left(\frac{\sigma^2}{\sigma^2} \frac{S_Y^2}{S_X^2} > 2\right) \\ &= P\left(\frac{S_Y^2/\sigma^2}{S_X^2/\sigma^2} > 2\right) \\ &= P\left(\frac{\frac{m-1}{\sigma^2} \frac{S_Y^2}{m-1}}{\frac{n-1}{\sigma^2} \frac{S_X^2}{n-1}} > 2\right) \\ &= P\left(\frac{\chi_{m-1}^2/(m-1)}{\chi_{n-1}^2/(n-1)} > 2\right) \\ &= P(F_{m-1, n-1} > 2) = P(F_{15,30} > 2) \end{aligned}$$


```
# pf = CDF of F ;  
  
# area to the left of 2;  
  
pf(2,15,30);  
  
## [1] 0.948209  
  
# area to the right of 2;  
  
1-pf(2,15,30);  
  
## [1] 0.051791
```